

Final Report

(CS 418 - Bhavisha Ahuja)

1. Introduction

This project analyzes socioeconomic disparities across four major U.S. metropolitan areas: Chicago, New York City, Dallas, and Oklahoma City. The initial objective was to model unemployment rates at the census tract level. However, unemployment showed insufficient variance across tracts to support meaningful statistical prediction. We therefore redirected our focus to poverty rate, which exhibited substantial variation and clearer socioeconomic patterns suitable for modeling.

Using census tract data from the American Community Survey (2019-2023), we examine which factors most effectively predict neighborhood-level poverty. The analysis compares multiple modeling approaches, from the linear regression methods covered in coursework to more complex ensemble learning techniques, to determine which framework best captures the relationships in our data.

2. Data Retrieval & Construction of the Dataset

Data was retrieved from the U.S. Census Bureau's American Community Survey 5-Year API (2019-2023 period). A custom Python script automated the data collection process, retrieving tract-level socioeconomic indicators for all counties within the four selected metropolitan areas. The script calculated several derived variables from the raw ACS data:

- Unemployment Rate: percentage of the civilian labor force currently unemployed
- Poverty Rate: percentage of the population with income below the federal poverty threshold
- Bachelor's+Rate: percentage of adults aged 25+ with at least a bachelor's degree
- Median Household Income: midpoint household income value for the tract
- Mean Travel Time: average commute duration in minutes

Data cleaning removed observations with invalid values, including negative incomes and mathematical errors in rate calculations. The final dataset contains 4,406 census tracts across the four metropolitan areas. All cleaned data was consolidated into a single file (`urban_economic_opportunity_data.csv`) for subsequent analysis.

After retrieval, the script performs preprocessing and derives key indicators:

Column Name	Description
GEOID	Unique 11-digit census tract identifier.
City	The major city and state assigned to the tract.
Unemployment_Rate	% of the civilian labor force that is unemployed.
Poverty_Rate	% of the population whose income is below the poverty level.
Bachelors_Plus_Rate	% of the population aged 25+ with a Bachelor's degree or higher.
Median_Household_Income	Midpoint income of households in the tract.
Mean_Travel_Time_Minutes	Average one-way commute time for workers in the tract.

The final cleaned dataset (urban_economic_opportunity_data.csv) contains all four cities in a unified format and is used for the EDA and modeling sections of the project.

3. Exploratory Data Analysis (EDA)

We began the EDA by inspecting the structure of the dataset, validating variable ranges, and removing invalid observations (e.g., negative household income, division-by-zero in rate calculations). After cleaning, the final dataset contained 4,406 census tracts across Chicago, New York City, Dallas, and Oklahoma City.

3.1 Summary Statistics

The first step was to generate descriptive statistics for the key socioeconomic indicators:

- Median Household Income
- Poverty Rate (%)
- Unemployment Rate (%)
- Bachelor's+ Rate (%)

- Mean Travel Time (minutes)

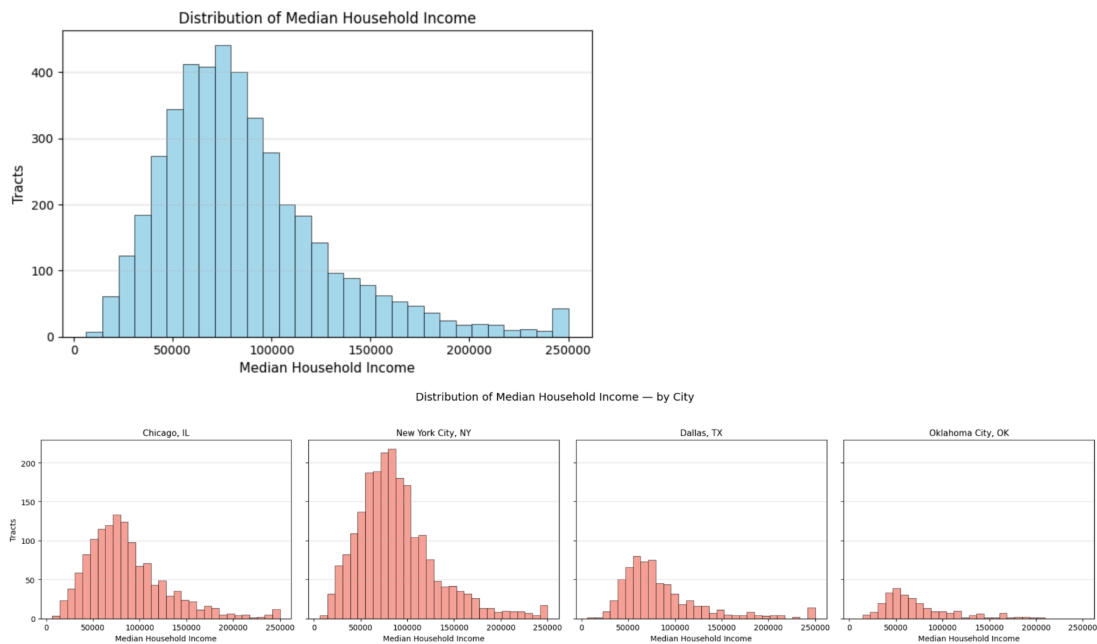
This provided a baseline understanding of central tendencies, variability, and potential outliers.

5]:

	GEOID	ACS_Year	Unemployment_Rate	Poverty_Rate	Bachelors_Plus_Rate	Median_Household_Income	Mean_Travel_Time_Minutes
count	4.406000e+03	4406.0	4406.000000	4406.000000	4406.000000	4406.000000	4406.000000
mean	3.236124e+10	2023.0	0.175374	15.558679	38.598566	86353.029732	26.105502
std	1.080262e+10	0.0	1.798468	11.450000	23.104880	42820.214101	13.104586
min	1.703101e+10	2023.0	0.000000	0.000000	0.270000	6311.000000	0.030000
25%	1.703182e+10	2023.0	0.000000	6.892500	20.292500	57317.750000	16.485000
50%	3.604707e+10	2023.0	0.000000	12.555000	33.840000	78185.500000	24.175000
75%	3.608111e+10	2023.0	0.000000	21.280000	53.047500	105034.750000	33.645000
max	4.811398e+10	2023.0	86.760000	76.990000	100.000000	250001.000000	106.930000

3.2 Distribution of Economic Indicators

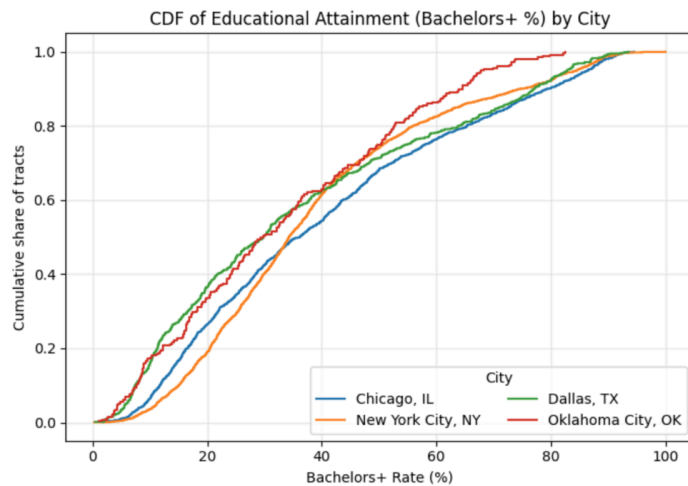
We analyzed the distribution of Median Household Income across cities and overall. Income displayed a right-skewed pattern with a long tail, aligning with typical U.S. household income distributions.



These visualizations demonstrated clear differences among cities—New York City tracts tended to have higher median incomes than Oklahoma City.

3.3 Education Levels (Bachelor's+ Rate)

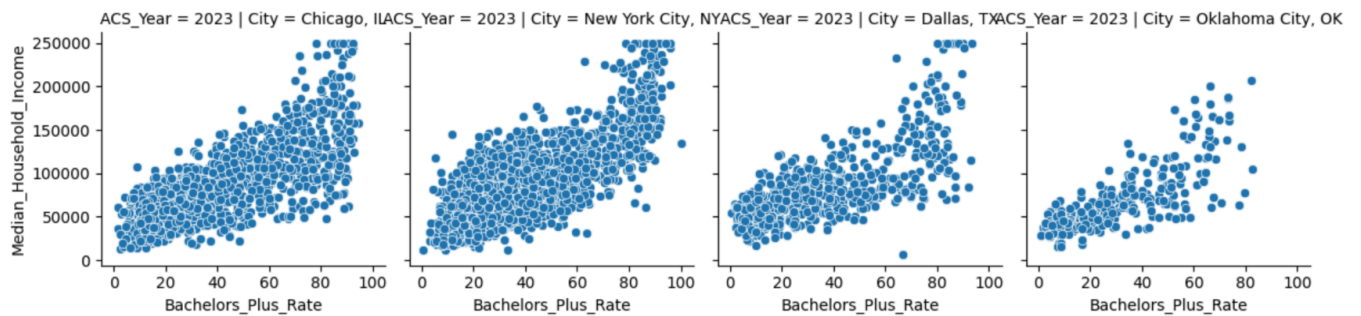
To evaluate variations in educational attainment, we plotted the Cumulative Distribution Function (CDF) of the *Bachelor's+ Rate* for each city.



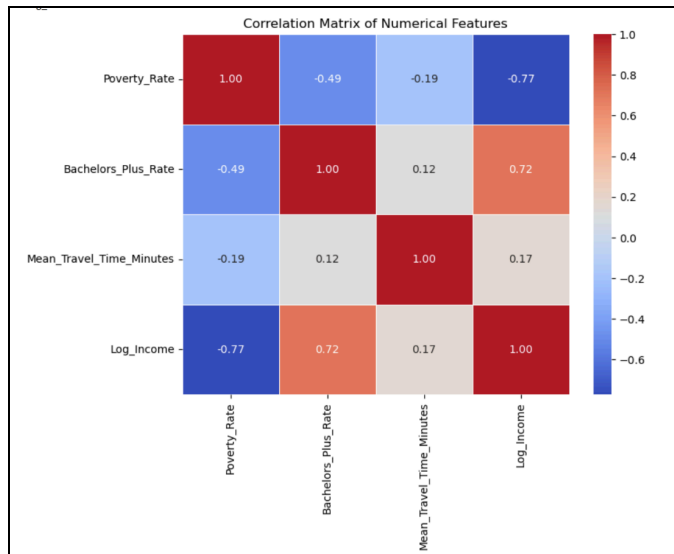
This revealed that New York City and Chicago have significantly higher concentrations of highly educated census tracts compared to Dallas and Oklahoma City

3.4 Income vs. Education Relationship

To explore the relationship between wealth and educational attainment, we used scatterplots to compare Bachelor's+ Rate with Median Household Income.

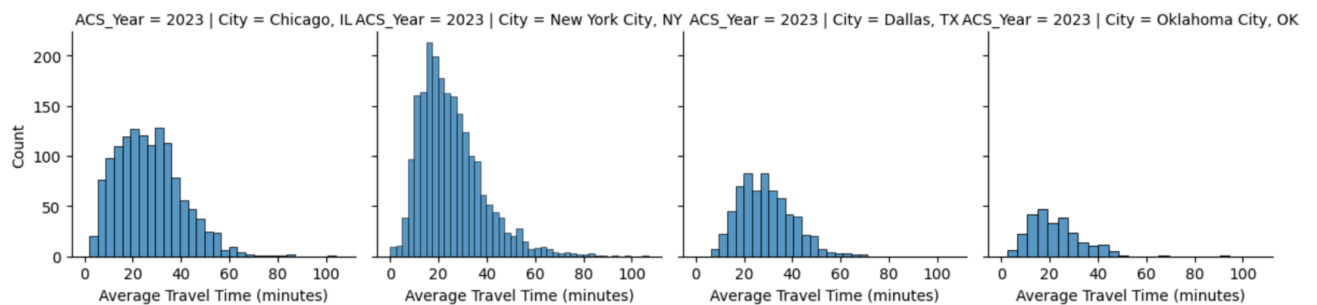


The positive correlation pattern suggests that areas with higher educational levels generally correspond to higher household incomes.

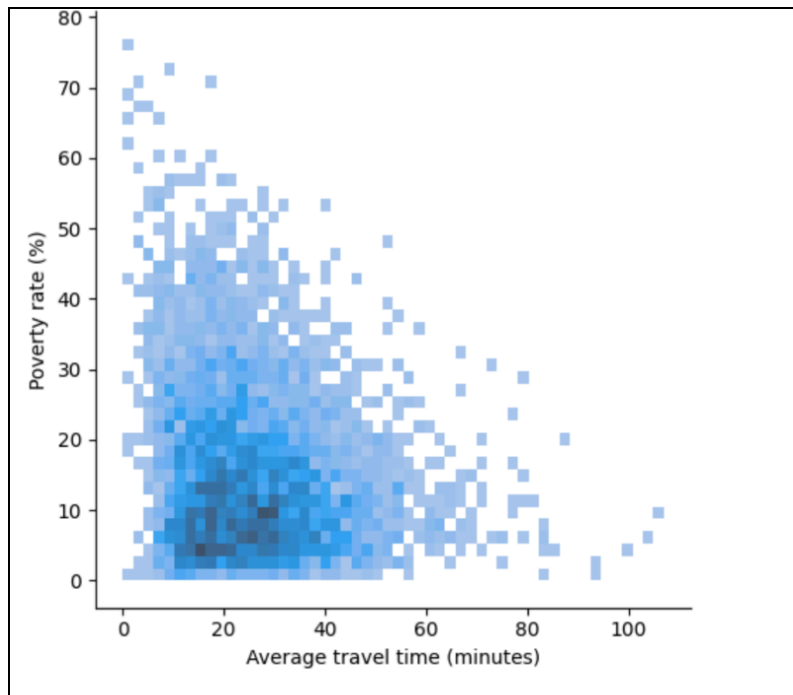


3.5 Travel Time Analysis

We examined the distribution of Mean Travel Time to Work across all tracts.



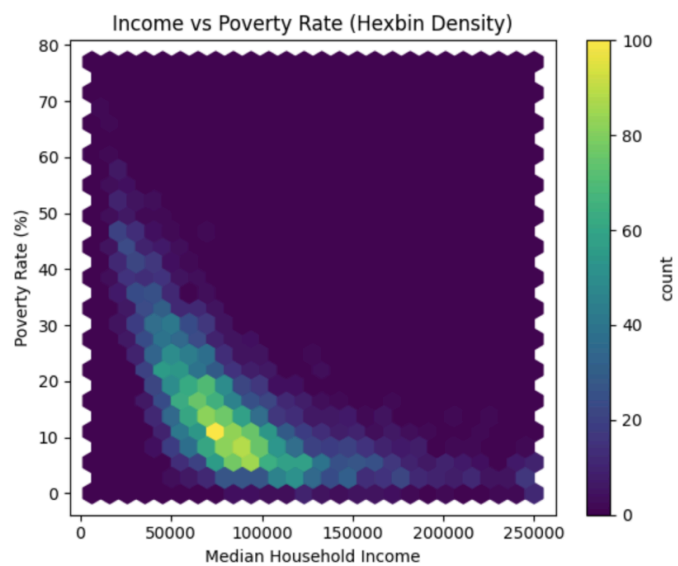
Most commute times fall between 20–40 minutes, consistent with metropolitan averages. We also explored the relationship between poverty and commute time using scatterplots.



The relationship is weak, suggesting that commute time alone does not directly explain poverty rate variations.

3.6 Poverty & Income Relationship

One of the most informative EDA visuals was the Income vs. Poverty Rate Hexbin Plot, which captures density and non-linear relationships.



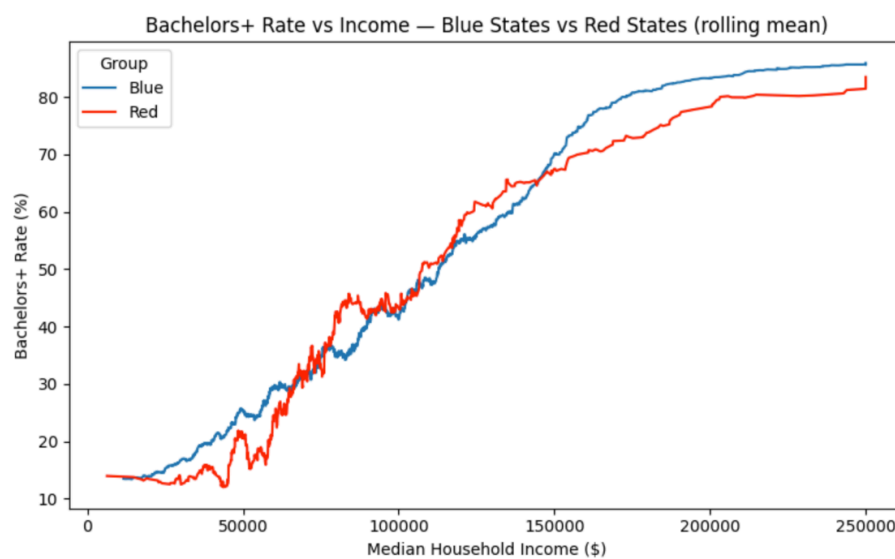
This chart clearly confirmed that lower-income census tracts show substantially higher poverty rates, and that poverty tends to be concentrated in the lower end of the income distribution.

3.7 Blue vs. Red Cities (Optional Hypothesis Test)

Your notebook also included a comparison grouping:

- Blue cities: Chicago & New York City
- Red cities: Dallas & Oklahoma City

Plotting rolling medians for income vs. education highlighted that "Blue" cities maintain higher Bachelor's+ rates at similar income levels.



This supports the idea that cities differ not just in economic outcomes but also in educational structure relative to income.

4. Model Training and Methods

Four regression models were implemented to predict poverty rate unemployment rate:

Multiple Linear Regression served as the baseline approach, consistent with course methodology. This model provides interpretable coefficients but assumes linear relationships between predictors and the outcome variable.

Random Forest Regressor constructs an ensemble of decision trees, each trained on bootstrapped samples of the data. This approach captures non-linear relationships and interaction effects while reducing overfitting through averaging.

XGBoost Regressor implements gradient boosting with regularization techniques that have proven effective in structured data prediction tasks. The algorithm builds trees sequentially, with each new tree correcting errors from previous iterations.

Gradient Boosting Regressor follows a similar sequential tree-building process but with different regularization and learning approaches compared to XGBoost, often yielding more stable predictions.

Evaluation Strategy Used for All Models:

- Train/test split
- 5-fold cross-validation
- R^2 (variance explained)
- MAE (average error in percentage points)

This ensures we compare both *accuracy* and *model stability* across methods.

5. Model Performance Comparison

Multiple Linear Regression (Baseline):

- Test R^2 : 0.5618
- MAE: 5.69 percentage points
- Cross-Validation: 0.5674 ± 0.0266

XGBoost Regressor:

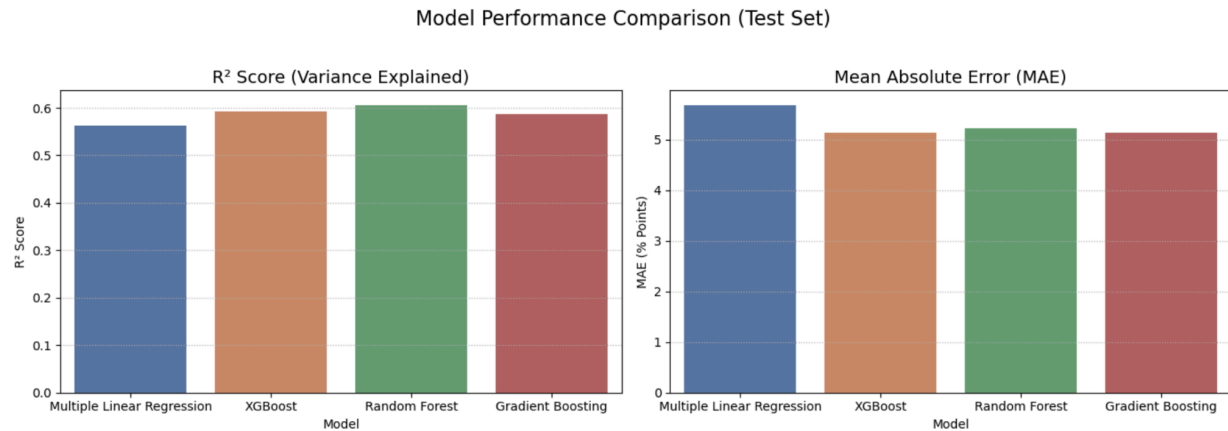
- Test R^2 : 0.5927
- MAE: 5.14 percentage points
- Cross-Validation: 0.6190 ± 0.0258

Random Forest Regressor (Best Performing):

- Test R^2 : 0.6065
- MAE: 5.22 percentage points
- Cross-Validation: 0.6096 ± 0.0241

Gradient Boosting Regressor:

- Test R^2 : 0.5870
- MAE: 5.14 percentage points
- Cross-Validation: 0.6235 ± 0.0281



Random Forest achieved the highest explanatory power on the test set, while Gradient Boosting demonstrated strong cross-validation consistency.

6. Comparison With Course Method (Linear Regression)

Multiple Linear Regression explained 56.2% of the variance in Poverty Rate, establishing a reasonable baseline.

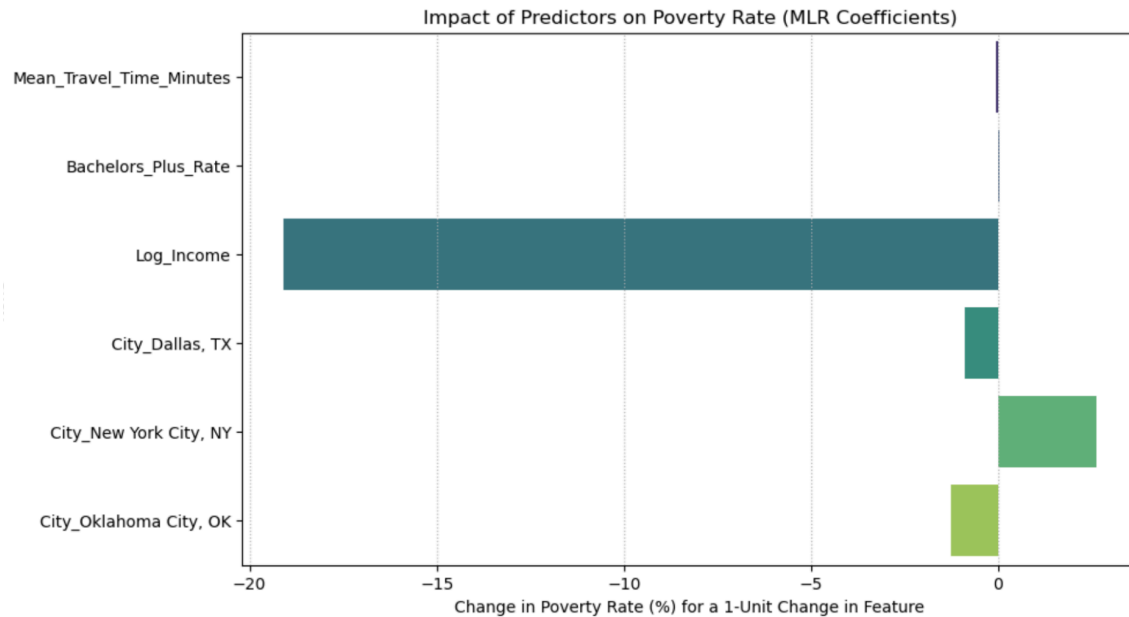
Random Forest improved this to 60.6%, showing a clear gain in predictive power.

This indicates: Poverty is shaped by *non-linear* interactions between variables (income, geography, education).

- Linear models cannot fully capture these complexities.
- Ensemble models—Random Forest, Gradient Boosting, XGBoost—are better suited for socioeconomic prediction tasks.

However, Linear Regression remains important because:

- It shows the *direction* of impact for each variable.
- The coefficients clearly highlight that income and city are dominant predictors.



7. Interpretation and Key Predictors

Across all models, a few predictors consistently stood out as the main drivers of Poverty Rate. The strongest and most stable predictor was log-transformed household income, which showed a large negative relationship with poverty—higher-income tracts almost always had lower poverty levels.

The city indicator variables also played an important role. Even after controlling for income, education, and travel time, poverty rates varied noticeably across the four metropolitan areas, suggesting that regional economic structure still influences outcomes.

Education, measured through the Bachelor's+ attainment rate, contributed moderately to the predictions. While the relationship was not as strong as income, higher educational attainment generally corresponded to lower poverty. Commute time showed a smaller, but still consistent, negative association.

The feature importance rankings from the ensemble models (Random Forest, XGBoost, and Gradient Boosting) supported these patterns. Income and geographic location dominated, with education and travel time serving as secondary predictors.

8. Conclusions

Overall, the analysis shows that poverty is shaped by several interacting socioeconomic factors rather than any single variable. While Linear Regression provided a useful baseline and offered clear interpretability, the ensemble models performed noticeably better, with Random Forest explaining roughly 60% of the variation in Poverty Rate across census tracts.

The improvement of ensemble models over linear regression suggests that poverty is influenced by non-linear relationships—for example, the interaction of income, education, and city-level conditions. Income emerged as the most important factor, but city effects and educational differences also contributed meaningfully.

These results highlight how combining traditional statistical approaches with modern machine-learning methods can provide a more complete understanding of urban economic inequality. Insights like these can help guide policymakers as they consider housing programs, education access, transportation planning, and targeted poverty-reduction strategies.

9. Possible Improvements

There are several ways the modeling could be improved in future work:

- Hyperparameter tuning for XGBoost, Random Forest, and Gradient Boosting could improve performance beyond the baseline models we used.
- Adding additional ACS indicators—such as housing cost burden, labor force participation, or rent-to-income ratios—would give a more complete socioeconomic picture.
- Incorporating spatial structure (e.g., tract adjacency or distance to job centers) could help capture geographic effects more accurately.
- Including interaction terms among predictors may reflect deeper relationships not captured by the current model.
- Developing a combined economic distress index could help summarize multiple dimensions of hardship into a single, interpretable metric.
- Using SHAP values or permutation importance would provide a clearer explanation of how the ensemble models make predictions.

These additions would make the model more accurate and also help policymakers better understand the factors contributing to economic inequality across cities.

10. Appendix: Individual Contribution Statement

Bhavisha Ahuja's Contribution:

1. Correlation Analysis & Feature Refinement

- Generated a full correlation matrix for all socioeconomic predictors.
- Identified low-correlation predictors and removed statistically weak variables to reduce noise.
- Cleaned, standardized, and finalized the feature set used across all models.

2. Model Development & Expansion

I significantly expanded the modeling section beyond the original baseline by implementing and evaluating multiple advanced models:

- Random Forest Regressor
- Gradient Boosting Regressor

This modeling extension strengthened the project and provided meaningful comparison to the course-required Linear Regression model.

3. Visualization Updates

I updated, corrected, and added several critical visualizations:

- Model Performance Comparison Chart (all four models)
- MLR Coefficient Plot (updated to reflect Poverty Rate, not Unemployment Rate)
- Residual Plot for best-performing model (Random Forest) with corrected axis labels
- Adjusted color schemes, titles, and readability of interpretation plots
- Ensured consistent labeling across all figures

These visuals are now included in both the notebook and presentation slides.

4. Analytical Rewrite & Interpretation

I fully rewrote the modeling explanation section to match the updated approach, including:

- Clear justification for switching the target from Unemployment Rate → Poverty Rate
- Comparative performance discussion across all four models
- Explanation of why ensemble models outperform linear regression
- Interpretation of key predictors (log-income, city effects, education, commute time)
- Assessment of model reliability (residual patterns, CV stability)

5. Recommendations & Future Work

I drafted the full “Possible Improvements” section to outline Hyperparameter tuning, Adding additional ACS socioeconomic indicators, Spatial modeling possibilities, Interaction effects, Economic distress index construction and SHAP / permutation importance for deeper model explainability.

6. Final Presentation Contribution

I drafted the presentation, which included designing the final presentation slides and discussing the script of the presentation with the group on discord.

Muhammad: Initial prediction model implementation and data retrieval.

Filip: EDA visualizations and early modeling support.

Rohit: Prediction variable discussion and modeling rationale.

Dima: EDA structure, integration of sections, and project coordination.